Week 2 Report: Multiple Regression and Inference on Airbnb Listings in Puglia

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Introduction

This report extends the analysis from Week 1 by introducing multiple regression, neighborhood-level controls generated via AI agents, and formal inference (t-tests, confidence intervals, F-tests) on Airbnb listings in Puglia.

Task 1: Data preparation and baseline models

Starting from the cleaned Puglia dataset, listings were filtered to those accommodating up to 6 guests. A standardized review score variable (*rating*, mean 0, sd 1) was created.

Two baseline models were estimated:

- **Model 1**: $\log(\text{price}) = \beta_0 + \beta_1 \text{rating} + u$
- **Model 2**: $\log(\text{price}) = \beta_0 + \beta_1 \text{rating} + \beta_2 \text{accommodates} + u$

The omitted-variable decomposition was performed by regressing *rating* on *accommodates* to obtain $\tilde{\delta}_1$. This confirms the OVB formula: $\tilde{\beta}_1 = \hat{\beta}_1 + \hat{\beta}_2 \tilde{\delta}_1$, where $\tilde{\beta}_1$ is the coefficient from Model 1, and $\hat{\beta}_1$, $\hat{\beta}_2$ are from Model 2.

Task 2: Neighborhood controls and full model

Only neighborhoods with at least 100 observations were retained (24 locations across Puglia). Four neighborhood characteristics were generated using AI agents:

- **Coolness**: attractiveness for the 20–30 age group and nightlife.
- **Centrality**: logistic/touristic importance and connectivity within Puglia.
- **Quietness**: night-time silence (higher = quieter).
- **Fanciness**: prestige, architectural elegance, and shop/restaurant quality.

Three AI agents were queried with the same prompt: ChatGPT, Gemini (thinking mode) and Perplexity (Deep Research). The final scores are a weighted average (ChatGPT 0.25, Gemini 0.25, Perplexity 0.50), giving more weight to Perplexity for its more detailed and comprehensive research process. All agent outputs (scores as `.csv`, reasoning as `.txt`) and the prompt are stored in `analysis/input/agents/`.

Table 1 reports the summary statistics for the four neighborhood variables.

Table 1: Summary Statistics for Neighborhood Variables

	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
Coolness_wavg	1	7339	8.05	1.18	8.62	8.16	0.56	4.88	9.88	5.00	-0.80	-0.59	0.01
Centrality_wavg	2	7339	7.97	1.36	8.25	8.05	1.85	4.75	9.88	5.12	-0.31	-0.53	0.02
Quietness_wavg	3	7339	5.40	1.61	5.25	5.33	2.41	2.62	8.25	5.62	0.31	-0.98	0.02
Fanciness_wavg	4	7339	7.95	0.88	8.00	7.97	1.30	6.12	9.12	3.00	-0.10	-1.18	0.01

A third model was estimated adding the four neighborhood controls:

- **Model 3**: $\log(\text{price}) = \beta_0 + \beta_1 \text{rating} + \beta_2 \text{accommodates} + \beta_3 \text{coolness} + \beta_4 \text{centrality} + \beta_5 \text{quietness} + \beta_6 \text{fanciness} + u$

Table 2 reports the regression results for all three specifications.

Table 2: Regression Results: Model 1, Model 2, and Model 3

	Log(price)-rating	Log(price)-rating-acc	Log(price)-final
(Intercept)	4.410***	4.006***	3.935***
	(0.005)	(0.014)	(0.181)
rating	0.110***	0.106***	0.105***
	(0.005)	(0.005)	(0.005)
accommodates		0.109***	0.124***
		(0.003)	(0.004)
Coolness_wavg			-0.111***
			(0.017)
Centrality_wavg			0.040***
			(0.006)
Quietness_wavg			-0.063***
			(0.015)
Fanciness_wavg			0.126***
			(0.011)
Num.Obs.	10688	10688	7339
R2	0.047	0.129	0.187
R2 Adj.	0.047	0.129	0.187
AIC	109685.4	108727.2	74951.5
BIC	109707.2	108756.3	75006.7
Log.Lik.	-7704.099	-7224.010	-4650.829
RMSE	0.50	0.48	0.46

- p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Interpretation of Model 3. Centrality and fanciness have positive and highly significant coefficients: a one-unit increase in fanciness is associated with a 12.6% price increase, while a one-unit increase in centrality corresponds to a 4% increase. Quietness has a negative sign, suggesting that quieter (more peripheral) neighborhoods command lower prices. Coolness is surprisingly negative (-11.1% per unit). This may reflect higher supply pressure: cooler neighborhoods attract more hosts, increasing competition and pushing prices down. Figure 1 confirms a positive relationship between coolness and the number of listings.

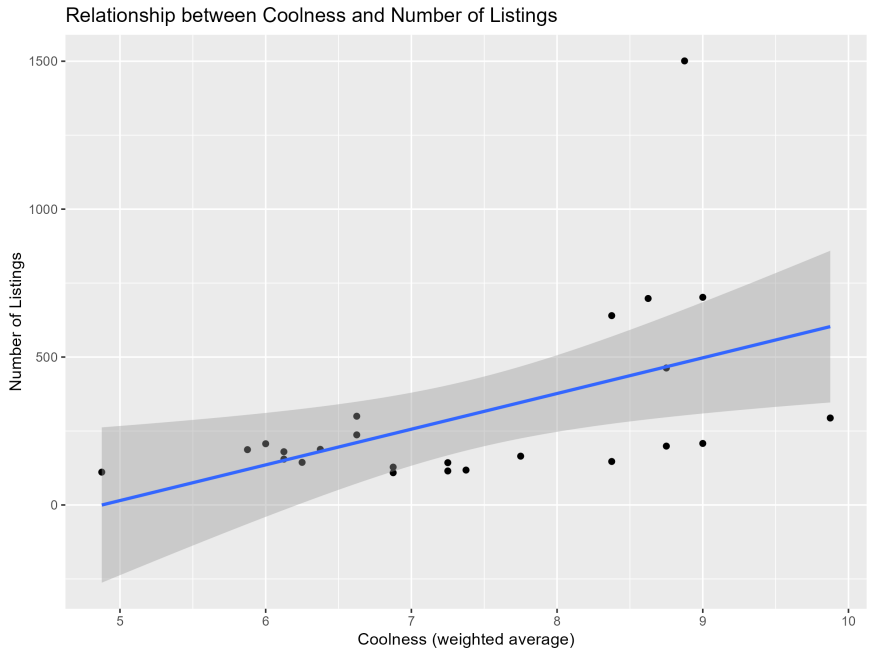

Figure 1: Relationship between Coolness and Number of Listings

Compared to Model 2, the *rating* coefficient is essentially unchanged, while *accommodates* rose from 0.109 to 0.124, suggesting that guest capacity becomes more important once neighborhood characteristics are controlled for. R^2 increased from 0.129 to 0.187 and RMSE decreased, confirming Model 3 as the best-fitting specification.

Task 3: Normality and t-tests

Residuals from Model 3 were extracted to assess normality. Figure 2 shows the residual density overlaid with a normal curve; the distribution exhibits heavier tails and slight skewness. The Jarque–Bera test formally rejects normality ($p < 0.05$), meaning we cannot assume normally distributed errors. However, given the large sample size ($n = 7{,}339$), inference based on asymptotic normality of OLS estimators remains valid by the CLT.

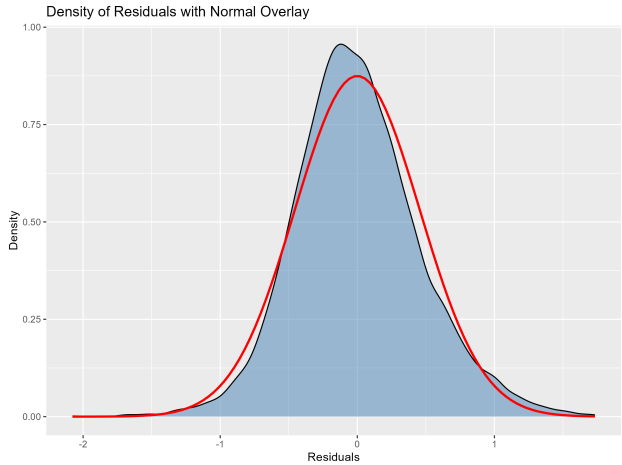

Figure 2: Density of Residuals with Normal Overlay

For the centrality coefficient, a two-sided t-test of $H_0 : \beta_{\text{centrality}} = 0$ vs. $H_1 : \beta_{\text{centrality}} \neq 0$ was conducted manually:

- **t-statistic**: 6.67
- **p-value**: 2.83×10^{-11}
- **5% critical value**: ± 1.96

The t-statistic falls deep into the rejection region. We reject H_0 and conclude that centrality is a statistically significant predictor of price. Figure 3 illustrates the t-distribution with rejection regions and the test statistic.

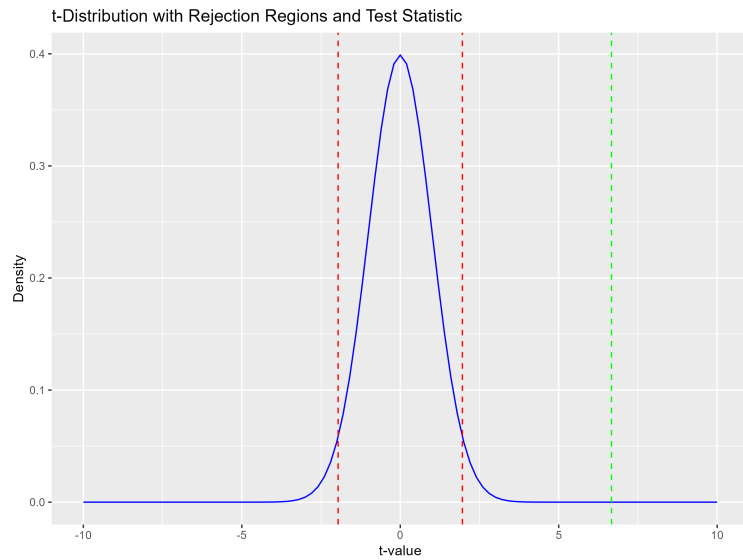

Figure 3: t-Distribution with Rejection Regions and Test Statistic

Task 4: Confidence interval and joint F-test

A 95% confidence interval for $\beta_{\text{centrality}}$ was computed manually as $\hat{\beta} \pm t_{0.975} \cdot \text{SE}(\hat{\beta})$. The interval does not include zero, which is consistent with the t-test rejection of H_0. Both approaches are equivalent: the t-test rejects at 5% if and only if the 95% confidence interval excludes the null value.

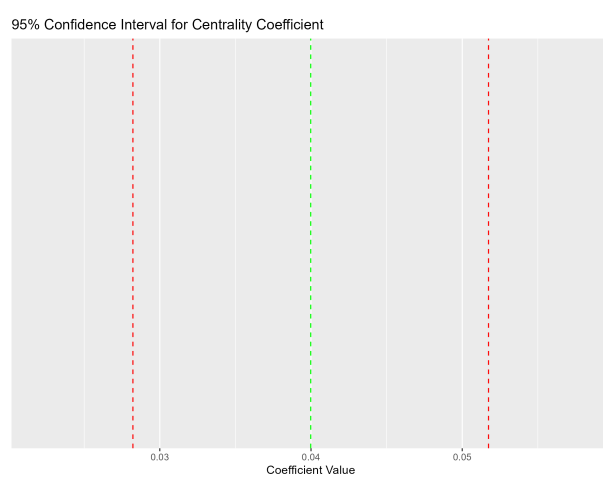

Figure 4: 95% Confidence Interval for the Centrality Coefficient

A joint F-test was conducted on the null hypothesis $H_0 : \beta_{\text{centrality}} = \beta_{\text{quietness}} = \beta_{\text{coolness}} = \beta_{\text{fanciness}} = 0$. The F-statistic was computed manually from the restricted (Model 2) and unrestricted (Model 3) residual sum of squares, and verified via `anova()`. The test strongly rejects the null, confirming that the four neighborhood characteristics are jointly significant predictors of log-price.